Determinants

- 2 by 2 determinants

(*) For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $ad-bc$.

For singular matrix, the determinant is zero.

(*) Row exchange reverse signs. $PA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & b \\ a & d \end{bmatrix}$ has $\det PA = bc - ad = -\det A$

(*) For 2 by 2 matrix. there would be 2 terms

For 3 by 3 matrix, there would be 6 terms

For n by n matrix, there would be $n!$ terms

- 3 by 3 determinants

Before we try to determine the determinant of a 3 by 3 matrix, let us try to understand how row swaps affect determinants:

i) Single row swap → sign change

(*) Swapping one pair of rows multiplies the determinant by -1.

$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, swap first two rows, $I' = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

↳ $\det = +1$, $\det(A) = +d$ ↳ $\det = -1$, $\det(A) = -d$

ii) Swapping two pairs → No sign change.

$I'' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ ⇝ $\det = (-1)(-1) = +1$

$\therefore \det(A) = +d$

iii) Swapping three pairs $\Rightarrow$ Sign change again

$$I''' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \Rightarrow \det = (-1)(-1)(-1) = -1$$

$\therefore \det(A) = -d$

* Swapping rows an even number of times keeps the determinant the same.

* Swapping rows an odd number of times negates the determinant.

So, for a matrix,

$$A = \begin{bmatrix} a & b & c \\ p & q & r \\ x & y & z \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} a & & \\ & q & \\ & & z \end{bmatrix} \quad \begin{bmatrix} & b & \\ p & & \\ & & z \end{bmatrix} \quad \begin{bmatrix} & b & \\ & & r \\ x & & \end{bmatrix}$$

$\det = +aqz$ $\quad$ $\det = -bpz$ $\quad$ $\det = +brx$

$$\begin{bmatrix} & & c \\ & q & \\ x & & \end{bmatrix} \quad \begin{bmatrix} & & c \\ p & & \\ & y & \end{bmatrix} \quad \begin{bmatrix} a & & \\ & & r \\ & y & \end{bmatrix}$$

$\det = -cqx$ $\quad$ $\det = +cpy$ $\quad$ $\det = -ary$

So, $\det(A) = \left[\begin{array}{ccc|cc} a & b & c & a & b \\ p & q & r & p & q \\ x & y & z & x & y \end{array}\right]$

Odd Row Swaps $\quad - \quad - \quad - \quad + \quad + \quad + \quad$ Even Row Swaps

$$= +aqz + brx + cpy - ary - bpz - cpx$$

* Here's a powerful insight:

To find the determinant of a 3X3 matrix, we needed six permutation matrix. Each of the permutation matrix contributes to one term.

... each term actually involves one number

One more thing, each term actually involves one number from every row and every column.

$\hookrightarrow$ From here, we can prove $n \times n$ matrix has $n!$ terms.

Cofactors and Formula for A^{-1}.

Here is the definition of cofactors C_{ij} and the $(-1)^{i+j}$ rule for their plus and minus signs.

For the i, j cofactor C_{ij}, **remove row i and column j** from the matrix A.
C_{ij} equals $(-1)^{i+j}$ times the determinant of the remaining matrix (size $n-1$).
The cofactor formula along row i is $\det A = a_{i1}C_{i1} + \cdots + a_{in}C_{in}$

(5)

$\hookrightarrow$ So for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, cofactor matrix, $C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$

Now, if we $AC^T = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = \begin{bmatrix} \det A & 0 \\ 0 & \det A \end{bmatrix} = (\det A)\, I$

$\Rightarrow AC^T = (\det A)$

$\Rightarrow A^{-1} = \dfrac{C^T}{\det A}$ [det A cannot be zero, for a matrix to have it's inverse]